

Hazel Lee

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Open to semiconductor roles starting May–September 2027 · Eligible to work in Canada · Open to relocation

EDUCATION

University of Waterloo, BSc Honours Nanotechnology Engineering (Co-op), Waterloo, ON

Expected April 2027

EXPERIENCE

SunDensity Inc., Research and Technical Assistant (co-op), Toronto, ON

Jan–Apr 2026

- Prepared thin plastic ITO-coated substrates, the base for the company's flexible solar cells, by cleaning them with various solvents and ozone.
- Prepared post-treatment solutions in a glove box, intended to improve adhesion and film quality.
- Monitored flow rate, coating speed, substrate temperature, and drying conditions during slot-die coating, aimed at producing consistent thin films.
- Inspected samples through optical inspection and microscope imaging to check film coverage and defects.

FortisBC, Engineering Co-op, T&D Standards (8 months), Trail, BC

Sep 2024–Apr 2025

- Developed 100+ bills of materials (BOMs) to create standardized design options for designers and technicians.
- Led weekly Monday online meetings between the T&D engineering and procurement teams, organizing the information and discussion points both sides needed.
- Evaluated three utility pole hole fillers selected by the supervisor to address costly woodpecker damage to transmission poles, and joined two operations crew members on a helicopter field test, documenting each product's pros, cons, and crew feedback.
- Findings informed the selection of the final filler, which became a recurring purchase for the company as a lower-cost alternative to replacing damaged poles.

University of Waterloo, Organic Electronics Lab, Laboratory Assistant (co-op), Waterloo, ON

Jan–Apr 2024

- Performed chemical synthesis regularly in the lab.
- Purified compounds independently, mainly by filtration and, a few times, column chromatography.
- Reproduced an experiment from a research paper together with the supervisor, recalculating ratios and molar masses for material weights that differed from the paper and having the calculations reviewed by the supervisor.

m-Health Solutions Inc., Data Associate (co-op), Hamilton, ON

May–Aug 2023

- Reviewed handwritten physician prescriptions and entered a high daily volume of prescription information through the company's website.
- Clarified unclear handwriting and unfamiliar medical terms with supervisors and colleagues before entering data, rather than relying on assumptions.
- Processed at least 50% more prescriptions than the previous co-op student in the role, based on a one-month comparison of the company's per-member completion counts.

ACADEMIC LAB

NE 340L Microfabrication lab, University of Waterloo

May–Aug 2025

- Built MIS capacitors with my lab group: PECVD, photolithography and etching.
- Compared one silicon-nitride film measured three ways (color chart ~1,125 Å, ellipsometry 1,231.89 Å, DEKTAK 1,849.57 Å) and explained in my individual report why I trusted the non-contact ellipsometry for the later calculations.

PROJECTS

Wafer-map yield visualizer, Browser tool

[demo](#)

- Turns die-level CSV data into a wafer map, yield summary, fail-bin Pareto and edge-versus-center comparison. The sample data are synthetic; I built it with AI assistance and checked its behavior by hand against test cases.

Semiconductor industry data exploration, Colab notebook

[code](#)

- Cleaned a public Kaggle dataset of semiconductor-company financials with Python and pandas and compared companies with Matplotlib charts.

SKILLS

Fabrication (hands-on, lab group): PECVD, photolithography, RIE, Al wet etch (NE 340L).

Thin-film and lab work: substrate cleaning (solvents, ozone), glove-box solution preparation, slot-die coating monitoring, optical and microscope inspection, chemical synthesis, purification (filtration, column chromatography), rotary evaporator, titration.

Measurement and analysis: film-thickness comparison (ellipsometry, DEKTAK, color chart), C-V/I-V data analysis in an individual lab report, Excel calculations.

Engineering documentation: bills of materials (100+), Excel tracking, requests for quotation, SAP MM (routine tasks), AutoCAD (basic).

Data and languages: Python basics (pandas, Matplotlib); English, Korean.